

SUTARIYA VEDANT JAMANBHAI

B.Tech ICT (Major) | Robotics (Minor) • 4th Year • PDEU, Gandhinagar

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EDUCATION

Pandit Deendayal Energy University (PDEU)

Aug 2023 – May 2027

B.Tech in Information and Communication Technology | Minor in **Robotics** | Student ID: 23BIT209

CPI: **8.67** | Gandhinagar, Gujarat

TECHNICAL SKILLS

Languages: Python, C, Java, MATLAB

Frameworks & Libraries: Flask, Django, FastAPI, LangChain, Ollama, Scikit-learn, TensorFlow, Keras, OpenCV, ROS2 Jazzy Jalisco, micro-ROS

Embedded & Hardware: ESP32, Raspberry Pi 4/5, AVR 8085, I2C / SPI / GPIO / UART, DHT11, ADXL345, KX134-1211, TG30 LiDAR, OpenManipulator-X

AI / ML / LLM: RAG Systems (FAISS, LlamaIndex), LLM Fine-tuning (LLaMA-2, QLoRA), Agentic AI, Computer Vision, Signal Processing

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Nginx, GitHub Actions, MongoDB Atlas

Web & Databases: Next.js, React, Vite, HTML/CSS, SQL, MongoDB, Firebase, Firestore

Protocols & Concepts: HTTP/REST, MQTT, SLAM, Odometry, MCP (Model Context Protocol), FreeRTOS

PROJECTS

PhysAgent — Embodied AI Agent Platform

2024 – Present

Raspberry Pi 5 + ESP32 Nodes | MCP | ROS2 | Python

- Designed a 10-stage build plan (S0–S9) for an AI agent that maps LLM reasoning to real-world GPIO, I2C, and SPI commands on Raspberry Pi 5.
- Built a manifest.json-driven Node Registry enabling auto-generation of MCP tool definitions for connected hardware peripherals.
- Integrated micro-ROS and ROS2 Jazzy Jalisco on Ubuntu 24.04 for sensor fusion, autonomous navigation, and manipulator control.
- Demonstrates full-stack AI-hardware bridging: natural language → agentic plan → physical actuation.

FactoryMind AI — Industrial Copilot

2024 – Present

ESP32 | Raspberry Pi 5 | LLM | GPIO / PLC | Python | FastAPI

- Building an edge-cloud SaaS platform that translates natural language to GPIO/PLC control signals for industrial automation.
- Architected a hybrid edge-cloud pipeline: ESP32 nodes capture sensor data; edge Raspberry Pi runs local LLM inference; cloud handles dashboarding and fleet management.

- Designed roadmap targeting Indian SME manufacturing sector with global expansion potential.

Bearing Fault Classification System

2024

Signal Processing | ML | RAG | LLaMA-2 | QLoRA | Windows App

- End-to-end fault detection system for bearing health (HB/ORD/IRD classes) using vibration data from KX134-1211 accelerometer via Raspberry Pi 4.
- Extracted time-domain features (RMS, Peak-to-Peak, Kurtosis, Crest Factor, Form Factor) and trained classification models; results exported to Excel.
- Implemented 5 RAG variants (feature-level, document-level, hybrid, few-shot, rule-based) using OpenAI API + FAISS for LLM-driven fault explanations.
- Fine-tuned LLaMA-2 with QLoRA on RTX 3060 (6 GB VRAM) on Windows 11 for domain-specific diagnostic text generation.
- Built Windows desktop app with wireless CSV transfer and integrated LLM explanations — patent-oriented project.

YatraAI — AI-Powered Travel Planning Platform

2024

AWS | Docker | FastAPI | Next.js | MongoDB Atlas | Google Gemini 1.5 Pro

- Developed a full-stack cloud-native travel planner with a 3-tier user system (Guest / Free / Premium) deployed on AWS EC2 with S3 storage.
- Integrated Google Gemini 1.5 Pro for itinerary generation; FastAPI backend containerized with Docker; MongoDB Atlas as primary database.
- Designed multi-stage document generation workflows and architecture diagrams for academic evaluation under Dr. Deepak Sharma, PDEU.

GitHub: github.com/Vedant28082005/travel-ai-planner-aws

IoT + LLM Environment Monitor

2024

ESP32 | DHT11 | DeepSeek-R1 | LangChain | Ollama | Flask

- Recorded live Temperature and Humidity from a hostel room via ESP32 + DHT11; transmitted to Flask API server via HTTP every 15 seconds.
- Pipelined sensor data into DeepSeek-R1 (via LangChain + Ollama) to auto-generate natural language environment reports; extended to DeepSeek LLM for agricultural insights.

RAG PDF Assistant

2023

LangChain | Ollama | FAISS | LLaMA-3 3B | Distilbert

- Built a local PDF question-answering system using LLaMA-3 (3B params) as the generator and Distilbert-base-uncased as tokenizer, with FAISS vector store.
- Supports upload-and-query workflow; usable as a personal document assistant or domain-specific LLM knowledge base.

Autonomous Logistics Robot

2024

ROS2 Jazzy | TG30 LiDAR | SLAM | micro-ROS | OpenManipulator-X

- Assembled and wired OpenManipulator-X robotic arm; configured ROS2 nodes for SLAM-based mapping using TG30 LiDAR on Ubuntu 24.04.
- Implemented odometry-based localization and motor control via PWMA/PWMB pin configurations on a custom robotic chassis.

ACHIEVEMENTS & ACTIVITIES

- 🏆 Won college Hackathon — IoT + ML integrated project (vibration-based bearing fault detection).

-  Pursuing patent for Bearing Fault Classification system combining Signal Processing, ML, RAG, and LLM fine-tuning.
-  Coursework: Digital Signal Processing (Butterworth/Chebyshev filters), DBMS, Microprocessors (8085), Discrete Mathematics, Cloud Architecture, Network Algorithms.
-  Playing drums and music; strong interest in algorithmic trading (Nifty 50 backtesting, candlestick pattern analysis).

CERTIFICATIONS

Machine Learning A-Z: AI, Python

Udemy

December 2023

Deep Learning and Computer Vision A-Z

Udemy

February 2024

B.Tech ICT + Robotics Minor • PDEU • Graduating 2027 • Open to internships in AI, Embedded Systems & Cloud